ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

POWER ELECTRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 3rd year final examination: power semiconductor devices and losses, heat sinks, gate drives and snubbers, controlled and uncontrolled rectifiers, DC-DC converters (buck, boost, buck-boost, H-bridge), and inverters for motor drives.

Written by Academic Central for practice, following power electronics modules at South African universities (e.g. University of Pretoria EDF 320 Power Electronics: power semiconductor devices, heat sinks, snubbers and gate drives, AC-DC and DC-DC converters). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae and substitutions. Give answers correct to TWO decimal places, with units.
3. Assume ideal switches and diodes unless losses are asked for. The mains supply is 230 V RMS, 50 Hz (single-phase) or 400 V line, 50 Hz (three-phase).
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: POWER SEMICONDUCTOR DEVICES

1.1 Compare the power diode, the SCR (thyristor), the power MOSFET and the IGBT under the headings: how they are turned on and off, switching speed, and a typical application. (8)

1.2 How is an SCR turned on, and how is it turned off in an AC circuit? (3)

1.3 A MOSFET in a motor drive carries 10 A and has $R_{DS(on)} = 0.05\ \Omega$. Its switching energy (turn-on plus turn-off) is 0.2 mJ per cycle at 20 kHz.

1.3.1 Calculate the conduction loss. (2)

1.3.2 Calculate the switching loss and the total loss. (3)

1.4 For the MOSFET in 1.3: $T_{j(max)} = 150\ °C$, the ambient temperature is 40 °C, $\theta_{jc} = 1.0\ °C/W$ and $\theta_{cs} = 0.5\ °C/W$. Calculate the largest heat-sink thermal resistance θ_{sa} that can be used. (4)

[20]

QUESTION 2: GATE DRIVES AND SNUBBERS

2.1 Why does a power MOSFET need a gate-driver IC instead of being driven straight from a microcontroller pin? Mention the problem of the high-side switch. (4)

2.2 A MOSFET has a total gate charge of 60 nC. Calculate the average gate current needed to switch it on in 100 ns. (2)

2.3 What is the purpose of an RC snubber, and of a freewheeling diode, in a switching circuit? (4)

[10]

QUESTION 3: RECTIFIERS (AC TO DC)

3.1 A single-phase half-wave controlled rectifier feeds a resistive load from the 230 V supply with a firing angle of 60°.

3.1.1 Calculate the average output voltage. (4)

3.1.2 Sketch the output voltage waveform and define the firing angle. (4)

3.2 A single-phase fully controlled bridge feeds a highly inductive load (continuous current) of 10 Ω resistance from the 230 V supply at α = 30°.

3.2.1 Calculate the average output voltage and current. (5)

3.2.2 What happens when α is more than 90°, and where is this used? (2)

3.3 A three-phase diode bridge is fed from the 400 V supply. Calculate the average DC output voltage and state the ripple frequency. (4)

3.4 A three-phase fully controlled bridge is fed from the 400 V supply at α = 45°. Calculate the average output voltage. (3)

3.5 Explain why a rectifier with a large smoothing capacitor draws a poor-quality current from the mains. (3)

[25]

QUESTION 4: DC-DC CONVERTERS

FIGURE 1 shows a buck converter switching at 50 kHz. Assume continuous conduction and ideal components.

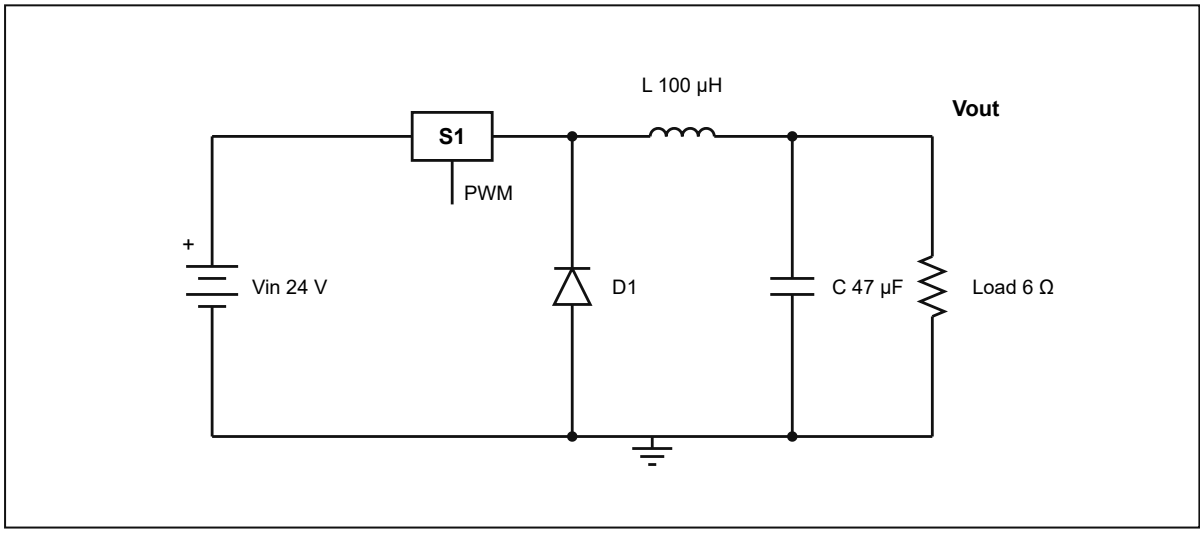


FIGURE 1: Buck converter

4.1 For an output of 12 V:

4.1.1 Calculate the duty cycle. (2)

4.1.2 Calculate the peak-to-peak inductor ripple current. (3)

4.1.3 Calculate the maximum and minimum inductor currents, and confirm the converter is in continuous conduction. (3)

4.1.4 Calculate the peak-to-peak output voltage ripple. (3)

4.2 Explain the difference between continuous and discontinuous conduction mode. (3)

4.3 A boost converter raises 12 V to 36 V. Calculate the duty cycle and the input current when the load draws 1 A. (4)

4.4 A buck-boost converter runs from 12 V at a duty cycle of 0.6. Calculate the output voltage. (2)

4.5 A linear regulator is used instead of the buck converter in 4.1 (24 V in, 12 V at 2 A out). Calculate its power loss and efficiency, and explain why switch-mode converters are used in battery robots. (4)

4.6 Draw an H-bridge driving a DC motor and explain how it gives forward, reverse and speed control. What is shoot-through, and how is it prevented? (6)

[30]

QUESTION 5: INVERTERS AND MOTOR DRIVES

5.1 A single-phase full-bridge inverter makes a square-wave output from a 48 V DC supply. Calculate the RMS output voltage and the RMS value of its fundamental. (3)

5.2 Explain sinusoidal PWM, and calculate the peak fundamental output of a full-bridge inverter with V_dc = 400 V and a modulation index of 0.8. (3)

5.3 Draw a block diagram of a variable-speed drive for an induction motor and explain each block. (3)

5.4 What is dead time in an inverter leg, and why is it needed? (2)

5.5 Why is the output of a square-wave inverter filtered, or replaced by PWM, before it drives a motor? (2)

5.6 When a motor on a variable-speed drive brakes, the DC-link voltage rises. Explain why, and one way to deal with it. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER